STI College Naga
Central Business District II, Brgy. Triangulo, Naga City
BSIT Department

Senior / Capstone Project
Capsule Proposal

Title:	***OneCaroyroyan: A Web-Based Barangay E-Service and Management Information System for Barangay Caroyroyan***
Proponent(s)/Researcher(s):	***John Michael V. Mujar*** ***Mary Ann S. Lumbria*** ***Mary Joy C. Basiño***
Objectives of the Study ***Objectives should be Specific, Measurable, Achievable, Relevant, Time-bound***	**The main objective of this study is to design and develop a web-based barangay e-service and management information system that enhances operations by making them more orderly, systematic, and aligned with procedural standards**. Specifically, this study aims to: • To support walk-in transactions through a digital internal processing system. • To centralize resident records for better organization and protection. • To semi-automate certificate generation using standardized templates. • To improve financial documentation through structured digital recording. • To enhance monitoring of blotter and complaint cases through centralized records. • To generate automated reports and dashboards for barangay officials. • To support transparent and procedural barangay governance.
Significance of the Study ***What entities or groups will benefit from your study? Rank them according to who is most directly benefited***	**The result of this study will benefit the following:** • *Barangay Captain – The system will provide dashboards and reports that support informed decision-making and effective governance.* • *Barangay Secretary – The system will help in managing resident records, complaints, and official correspondence more efficiently.* • *Administrator – The system will assist in processing certificate requests faster and more accurately through semi-automation.* • *Treasurer – The system will improve monitoring and validation of financial transactions and reporting.*

	• *Accounting Clerk – The system will simplify payment recording and receipt generation, ensuring accurate financial tracking.* • *Residents – The system will improve the speed and accuracy of services such as certificate issuance.* • *Barangay Caroyroyan – The system will enhance organization, transparency, and overall operational efficiency.* • *Future Researchers – The study will serve as a reference for future development of barangay management systems.*
Scope of the Study ***What will be covered (client / timeline) by your study? What do you intend to produce / create? What tools / technologies will you use?***	**This study focuses on the design and development of a web-based barangay e-service and management information system for Barangay Caroyroyan.** *The system will cover:* • *Walk-in transaction support through internal digital processing* • *Centralized resident information management* • *Semi-automated certificate generation* • *Financial transaction recording (cash-based)* • *Blotter and complaint management with status tracking* • *Official correspondence and document tracking* • *Automated reports and dashboards* • *Role-based access for authorized barangay personnel* *The system will be accessible through web browsers and will be used only by authorized barangay staff.*
Limitations of the Study ***What are the <u>uncontrollable</u> factors that will limit your study?***	**The study will be limited to the following:** • *The system supports walk-in transactions only and does not allow online certificate requests.* • *Online submission is limited to official correspondence from external sectors only.* • *The system is intended only for Barangay Caroyroyan.* • *The system depends on proper staff training and stable internet connectivity.* • *Financial transactions are limited to cash-based recording only.*
Delimitations of the Study ***What constraints will you impose upon your project to make sure it can be completed within the given time?***	**The study is limited to:** • *Internal barangay operations and management processes* • *Authorized users such as Barangay Captain, Secretary, Treasurer, Accounting Clerk, and Administrator* • *System features related to records management, certificate processing, financial recording, and reporting* • *Data collected within the duration of the study*

Project Features ***What are the foreseen useful IT-based attributes of your proposed project?***	**The proposed system will include the following features:** • *Role-Based Access Control (RBAC)* • *User account management* • *Centralized resident information management* • *Semi-automated certificate generation* • *Payment recording and receipt generation* • *Blotter and complaint management system* • *Official correspondence and document tracking* • *Dashboard overview with real-time data* • *Automated reporting (PDF and printable)* • *Charts and analytics for decision-making*
Relevant Skills and Experiences **Briefly and concisely explain your technical and non-technical skills, as well as the relevant experience that make you qualified to carry out this project (Not more than 300 words)**	*The researchers have basic skills in programming, web development, and database management. They also have knowledge of the System Development Life Cycle (SDLC). The researchers understand how barangay operations work, especially in records management, certificate processing, and reporting.* *These skills are enough to support the development of the proposed system.*
Requested Adviser:	Mr. Mark Joeven S. Entico
Capstone Coordinator:	Mr. Lance Stephen L. Bronzal

Prepared by:

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